Phase 3: Project Design Phase

System & UI Design Document

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System Architecture Flow

The system follows a standard client-server Machine Learning architecture:

1. **User Interface:** User interacts with HTML templates (`home.html`, `input.html`, `output.html`).
2. **Flask Backend:** Routes HTTP POST requests from the form submission to `app.py`.
3. **Data Processing:** Form string inputs are cast to `int64`, converted to a Pandas DataFrame, and reordered using `model_columns.pkl`.
4. **Prediction Engine:** The processed DataFrame is passed to `smartlender_tuned_xgboost.pkl`.
5. **Response:** Predicted class (1 or 0) and probability percentage are rendered back to the user.

Data Flow Logic

When user data arrives via HTTP POST, it is in string format. The Flask app executes a dictionary comprehension to convert all values to integers. This ensures XGBoost receives the exact `int64` format it was trained on. The `model_columns.pkl` file is then used to align the DataFrame columns precisely, preventing feature mismatch errors.

User Interface Mockups

1. Home Page (`home.html`): A clean landing page displaying project details, the tech stack used, and a prominent green "Start Application" button that routes to the form.

2. Input Page (`input.html`): A structured HTML form containing dropdowns for categorical variables (mapped to their integer encodings, e.g., Male=1, Female=0) and number inputs for continuous variables like `ApplicantIncome` and `LoanAmount`.

3. Output Page (`output.html`): A focused result page displaying the loan status (APPROVED in green text / REJECTED in red text) and the model's confidence percentage.